

JESUS PEREZ BAZAROT

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SUMMARY

Fourth-year Computer Engineering student and AI/ML developer, expected graduation 06/2028. Sole author of a continual-learning benchmark paper under review at a NeurIPS 2026 workshop, with all 375 experimental runs, code, and data public. Hands-on across the stack: hand-written CUDA kernels and GPU model optimization in PyTorch, speech and multimodal pipelines, LLM evaluation, and production-facing FastAPI and React systems. Prior CTO and co-founder experience. Seeking machine learning research or engineering roles.

EDUCATION

University of Seville <i>B.Sc. in Computer Engineering, Computer Engineering specialization</i>	2023 - 06/2028 (expected) <i>Seville, Spain</i>
Beijing Institute of Technology (BIT) <i>Erasmus exchange, Computer Science and Big Data Analysis</i>	Sep 2025 - May 2026 <i>Beijing, China</i>
– Coursework: Machine Learning, Big Data Analysis, Digital Logic	

PUBLICATIONS

Perez Bazarot, J. **Forgetting in World Models Does Not Follow Task Distance**. Submitted to the CL4FMAgents workshop at NeurIPS 2026 (8 pages, non-archival, under review). Full 26-page version, code, and all result files: github.com/PersusUS/WorldModelsBenchmark

- Built a component-level continual-learning benchmark isolating the transition component of a world model: 375 runs over 3 environment families (MiniGrid, MuJoCo/Gymnasium, DeepMind Control) by 3 dynamics-distance levels by 5 methods (finetuning, EWC, infinite replay, progressive networks, and UG-MTM, a proposed uncertainty-gated mixture of transition models), up to 10 seeds per cell; same-seed reruns reproduce every metric bit-for-bit
- Result 1: the labelled task-distance axis carries no rank information about forgetting (Spearman rho = 0.00). Forgetting peaks at the intermediate level in all three families, and the effect survives a rerun at twice the training budget
- Result 2: forgetting concentrates in the encoder, where isolated metrics do not look. Finetuning degrades task-A reconstruction by 811x while the transition-level metric reports an apparent improvement, and EWC's Fisher information is exactly zero over the encoder by construction
- Both headline results are negative, including one that refutes my own proposed architecture; next archival target is CoLLAs 2027

EXPERIENCE

OrgaAI <i>AI/ML Developer</i>	Jun 2025 - Sep 2025 <i>Spain</i>
– Owned the audio processing and transcription pipeline of a confidential conversational AI product	
– Cut pipeline runtime by 10x by reusing existing dataset transcripts instead of running full WhisperX passes	
– Researched and prototyped a Slot Attention integration for model architecture improvements (under NDA)	
ByTheWay Carpool (startup) <i>Co-Founder and Co-CTO</i>	2024 - 2025 <i>Spain</i>
– Co-founded a carpooling platform for Loyola University students and co-led its technical architecture: rider-driver matching, trip coordination, and operational tooling	
Minerva Research Group, University of Seville <i>Research Intern</i>	2023 - 2024 <i>Seville, Spain</i>
– Selected as an internal student researcher in first year, an early-entry position normally offered to later-year students	
– Built machine learning classifiers over University of Cordoba gas-emission datasets from Iberian ham samples to separate feeding categories (cebo vs. bellota)	

NetKey (technology startup)

2023 - 2024

Chief Technology Officer

Seville, Spain

- Ran the full technical stack as sole engineer: automated order handling, client communications, and internal email workflows, removing most manual operations overhead
- Built and deployed the company website end to end, including structure, content, and CI/CD

Telefonica, innovaTE program

2023

Technology Speaker, NB-IoT and Sustainability

Seville, Spain

- Presented emerging technologies to enterprise and institutional audiences as part of the innovaTE event team, owning the sustainability track with product-focused talks on Narrowband IoT
- Followed up directly with interested company representatives on use cases and technical value

PROJECTS

Selective-scan CUDA kernels | *CUDA C++, PyTorch C++ extensions*

2026

- Implemented the selective state space recurrence $h[t] = a[t] h[t-1] + b[t]$ as five CUDA kernels, from a shared-memory Blelloch scan up to a single-pass decoupled look-back scan (Merrill and Garland, NVIDIA NVR-2016-002), the algorithm behind CUB's DeviceScan
- Fastest kernel sustains 79.6-91.5% of theoretical DRAM bandwidth on an RTX 4050 Laptop (192 GB/s): 54x-69x over the best pure-PyTorch baseline and 2.18x-2.48x over the three-pass version, against 2.33x predicted by a memory-traffic model written before the kernel existed
- Built the inter-block protocol on atomicAdd, __threadfence and volatile flag loads, claiming tiles by atomic counter so no tile waits on an unscheduled predecessor; a warp-parallel look-back over shuffle and ballot intrinsics cut chain latency by up to 1.80x at 256 tiles
- Verified with 107 correctness checks against a float64 reference and 630 determinism runs; of five performance predictions registered before measuring, three held and two were refuted, both documented as refuted in the repository

HybridMamba-11, OpenAI Parameter Golf (PR #1365) | *PyTorch, CUDA, torch.compile, Triton, DDP*

2026

- Designed HybridMamba-11, a hybrid Mamba state space model and Transformer architecture (11 layers) with U-Net skip connections, the first state space model submission in the competition
- Diagnosed the core bottleneck: Mamba CUDA kernels are untraceable under torch.compile fullgraph mode: roughly 300 training steps in the 600s budget against about 20,000 for top entries
- Replaced the sequential O(L) recurrence with a parallel associative scan (Hillis-Steele) in pure PyTorch: O(log L) depth, fullgraph compilable, 48x measured speedup on the optimized path
- Packaged the submission at 31.8M parameters and 13.6MB compressed against a 16MB cap, after removing 12.5M dead parameters and fixing compile, dtype, and state-dict compatibility issues; entry open and preliminary, pending H100 time to validate the fullgraph path

Perseo, personal AI assistant | *Tauri 2, Rust, React, Python, Ollama, Gemini Live API*

2023 - Present

- Built and shipped a multi-agent assistant that runs as a Windows service with a watchdog: it reads email and calendar, triages with a local model (Qwen3 4B on Ollama), and notifies by Telegram only what matters, sending counts and never message subjects
- Implemented a file-based memory agent over an Obsidian vault with append-only writes, and gated every irreversible action behind an explicit Telegram confirmation
- Built the real-time voice mode: dual AudioContext streaming at 16 and 24 kHz over a Gemini Live WebSocket with barge-in and wake-word detection, plus in-memory screen capture through xcap in a Rust backend with no cloud persistence
- Hardened the repository to 919 unit tests, 17 end-to-end verifiers, and green CI, with a mobile-first PWA control panel; an earlier audit found and fixed an API key shipped inside the binary and a 7.5 s Rust-to-Python bridge latency

Multilingual LLM benchmark | *Python, evaluation pipelines, LLM APIs*

2026

- Designed a 9-language benchmark comparing Chinese and Western frontier models (DeepSeek V3, Qwen-Max, Gemini 2.5 Pro, Mistral Large, Llama 3.3 70B) across closed and open-ended task types
- Scored closed tasks by accuracy and open tasks by ROUGE-L and BERTScore, with a three-model LLM jury for open-ended judgments

- Paused: extending coverage to further frontier models is blocked on research API credits

ClassTranscriber | *Python, PyTorch, CUDA, faster-whisper, pyannote.audio, Ollama* 2025

- Built a fully on-device CLI pipeline for multilingual classroom audio: speaker diarization with pyannote, speech recognition with faster-whisper large-v3, and transcript cleanup with a local Gemma model, using no cloud APIs
- Added real-time audio preprocessing (spectral noise reduction, amplitude normalization) to raise transcription quality on accented English speech
- Structured it as a modular 7-stage pipeline with pytest unit tests and mocking, and used it for real classes during the exchange at BIT

AutoLinkedIn | *React, TypeScript, FastAPI, WebSockets, Supabase, pgvector* 2025

- Built an end-to-end web app that runs real-time bidirectional voice interviews (PCM16 streaming over AudioWorklet and the Gemini Live API) and generates LinkedIn posts from them
- Designed a retrieval-augmented generation pipeline (FastAPI, Supabase pgvector, Gemini embeddings) that holds a consistent personal voice across sessions and outputs trilingual posts in Spanish, English, and Chinese
- Integrated LinkedIn OAuth 2.0 with a one-click publish flow from a single React dashboard

Magic OCR | *Expo, React Native, VisionCamera, Supabase* 2026

- Built a trading-card scanner that uploads no images: a 64-bit difference hash of the luminance plane matched by Hamming distance against an index of 1,749 illustrations, reaching 97.5% first-try accuracy

SKILLS

Programming languages: Python, C, C++, Java, TypeScript, JavaScript, SQL, Assembly

GPU and systems: CUDA C++ (shared memory, warp shuffles, atomics, memory fences, occupancy and bandwidth analysis), parallel scan algorithms, PyTorch C++ extensions

Machine learning: PyTorch, CUDA, torch.compile, Triton, distributed data parallel training, mixed precision (BF16), quantization, Transformers, state space models (Mamba), world models (RSSM), continual learning (EWC, replay, progressive networks), retrieval-augmented generation

Research and evaluation: benchmark design, ablation studies, seed and reproducibility protocols, LLM evaluation, ROUGE-L, BERTScore, pandas, NumPy, scikit-learn, matplotlib

Speech and multimodal: faster-whisper, WhisperX, pyannote.audio, noisereduce, Web Audio API, AudioWorklets, real-time streaming, screen and webcam capture

Software and infrastructure: FastAPI, React, Tauri 2, Rust, Vite, Tailwind CSS, WebSockets, REST APIs, PostgreSQL, pgvector, Supabase, SQLite, Docker, Git, CI/CD, pytest, Linux, Ollama

Languages: Spanish (native), English (C1, Trinity 2020), Chinese (conversational)

AWARDS AND ACTIVITIES

Wanted Entrepreneurship Competition 2023

First place out of 20 teams Spain

Provincial Debate Competition 2023

Finalist Spain

NASA Space Apps Challenge 2023

Participant, swarm systems and robotics challenges Spain

University of Seville Algorithmics Club 2023 - Present

Member Seville, Spain